ISHTIAQ AHMAD

Software Engineer · Backend Developer · Machine Learning Engineer
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PROFESSIONAL SUMMARY

Software Engineer with hands-on experience in Python backend development, REST API design, machine learning engineering, and ETL data pipelines. Deployed a YOLOv8 object detection system (94%+ accuracy) integrated with Django REST APIs across 200+ government schools. Proficient in SQL schema design, PostgreSQL, Docker, and computer vision (OpenCV). Top 1.5th percentile IT graduate nationally (NSCT 2026) · CGPA 3.50/4.00.

PROFESSIONAL EXPERIENCE

Punjab Monitoring & Implementation Unit (PMIU) · **Software Engineering Intern** Jun – Aug 2025 · Lahore

- Fine-tuned YOLOv8 object detection model (94%+ accuracy), automating student headcount across 200+ government schools and cutting manual reporting time by ~80%.
- Built Django REST APIs integrating the CV pipeline with a PostgreSQL database — implementing end-to-end ETL data transformation, validation, and structured storage.
- Monitored ML model performance (accuracy, precision, recall, F1-score) and iterated on retraining cycles to continuously improve prediction quality.

EDUCATION

BS Software Engineering · University of Management and Technology (UMT), Lahore Expected Feb 2027 · CGPA: 3.50 / 4.00

TECHNICAL SKILLS

Languages	Python · SQL · C++ · JavaScript · HTML/CSS
ML & AI	YOLOv8 · TensorFlow Lite · Computer Vision · Object Detection · Edge AI
Backend	Django · REST API Design · API Integration · Docker · Microservices
Data Engineering	ETL/ELT Pipelines · Data Transformation · Data Visualization · Pandas · NumPy
Databases	PostgreSQL · SQLite · SQL Schema Design · Query Optimization
Tools	Git · OpenCV · PyQt5 · Docker · Agile · SDLC · Version Control

ACHIEVEMENTS

NSCT 2026: 98.5th percentile (Top 1.5%) among IT graduates nationwide. **Competitive Programming:** 1st Place SoftCom 2026 · 1st Place techFiesta 2025 · Runner-Up TechVerse 2025. **LeetCode:** 150+ problems solved (DSA, DP, Graph Theory).

TECHNICAL PROJECTS

"The Judge" — Automated Code Evaluation Platform — *Django · Python · Docker · PostgreSQL · REST API · JavaScript*

- Designed normalized SQL schema and Docker-containerized execution engine for multi-language code evaluation; implemented full ETL pipeline from submission ingestion through structured reporting, reducing manual overhead by ~90%.
- Built Django REST API backend with concurrent sandboxed execution; integrated JS frontend for real-time result display and performance analytics dashboards.

Vision-Based Autonomous Car with Indoor Positioning System (FYP) — *Python · Raspberry Pi 5 · YOLOv8 · OpenCV · Edge AI*

- Deployed YOLOv8 object detection and lane-following algorithms on Raspberry Pi 5 (Edge AI); engineered a real-time telemetry dashboard streaming live sensor data, location coordinates, and CV model outputs.
- Developed Python data ingestion and logging pipeline for multi-channel sensor streams enabling real-time visualization and post-hoc performance analysis.

Smart City Management System — *Python · PyQt5 · SQL · Edge AI · IoT*

- Designed SQL data models and edge-AI pipelines monitoring urban metrics (traffic, service requests) from IoT feeds; built PyQt5 dashboard with ETL-based structured report generation.